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Comprehensive Report

Alkene, Boron-ate complex, Hydroboration, Macrolide, Total synthesis

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A Modular Synthesis of Antitumor Macrolide (–)-Lasonolide A†

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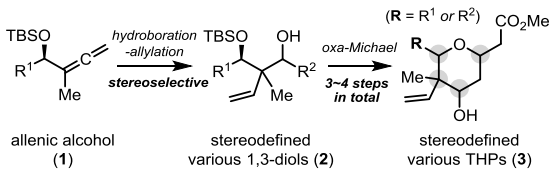
Summary of main observation and conclusion Lasonolide A was identified as a potent antitumor macrolide towards various cancer cell lines. The two tetrahydropyrans bearing multiple stereogenic centers as well as the polyene linkage attracted a dozen synthetic research groups to launch the total synthesis. Based on the synthetic methods developed in our group, namely, the hydroboration of allene and its subsequent allylation as well as the iterative hydroboration of allene and oxidation, the polyol subunits were efficiently constructed and then integrated into the final target. A new Julia olefination reagent, double-headed sulfone, was designed to promote the rapid coupling of two aldehydes bearing multiple functional groups to secure the whole carbon framework. Another highlight of our approach is the application of a traceless protecting group, 9-BBN (9-borabicyclo[3.3.1]nonane), to hide the secondary alcohol for debenzylation, and for the first time, to mask the carboxylic acid for Julia olefination under strong basic conditions.

Background and Originality Content

Marine natural products have emerged as a rich resource for drug discovery due to their fascinating structures and various biological activities. Since the first FDA-approved marine natural product-derived Ziconotide, an atypical analgesic agent for the amelioration of severe and chronic pain, dozens of marine natural products have been developed for clinical use.[1] Among the structurally diverse compounds isolated from the *deep sea*, polyketide plays a pivotal role in the development of marine drugs. A milestone in drug discovery is the development of Halaven® (eribulin mesylate), which is clinically used for the treatment of metastatic breast cancer.[2] Although this marketed drug is a truncation of the marine natural product halichondrin B, the formidable complexity of its chemical structure remains a synthetic challenge because the sample supply relies only on its total synthesis in the laboratory.[2b] To quickly provide sufficient materials of promising leads such as halichondrin B and embark on the medicinal studies of their derivatives, novel synthetic methods and strategies are still in high demand.

In the continuing interest in the field of polyketide synthesis, our laboratory has embarked on the development of novel methods and efficient strategies to assemble complex bioactive polyketides.[3] During the hydroboration of allenic alcohol **1** (Scheme 1), we noticed a selection of different alkylboranes would provide various 1,3-diols **2** in a highly diastereoselective manner after the allylation of the aldehydes, which were attributed to the stereodefined *E*- or *Z*-allyboranes derived from the hydroboration of allenes. Moreover, a generally found boratropic rearrangement ([1,3]-boratropic shift)[4] in allylboranes was largely suppressed due to the steric bulkiness of the alkyl substituents on boron. The superior reactivity of disiamylborane (Sia_2BH) to the alkene (the terminal olefin in allenes) and the slow 1,3-shift of boron facilitate the kinetic hydroboration-allylation to deliver an *anti,anti*-1,3-diol motif with an all-carbon quaternary center.[3c] Moreover, the subsequent transformations, including a bidirectional oxa-Michael cyclization, were able to produce diastereoisomers of tetrahydropyrans **3** in just 3—4 steps with excellent overall yields.

Scheme 1 Our previous approach to construct tetrahedropyrans (**3**)



The construction of a variety of tetrahydropyrans (THPs) bearing multiple stereocenters encouraged us to devise an efficient modular synthesis for lasonolide A (**4**), a marine polyketide featuring two characteristic THP rings. In 1994, McConnell and co-workers reported lasonolide A, a complex macrolide exhibiting a potent antitumor activity isolated from the Caribbean orange-red marine sponge, *Forcepia* sp. (Figure 1).[5] The originally proposed structure was later revised to be **5** on the basis of the

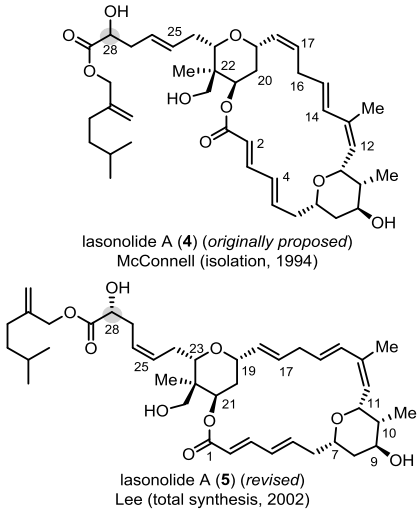


Figure 1 Structural revision of lasonolide A (**4** *vs* **5**).

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† Dedicated to the 70th Anniversary of Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences.

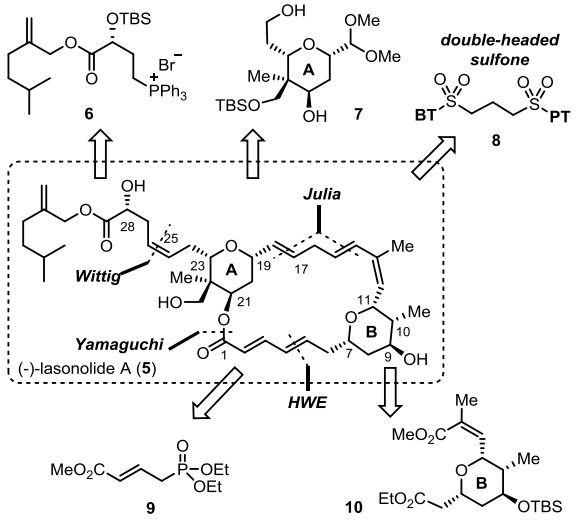
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collective syntheses of diastereoisomers by the Lee group.[6] Other lasonolide congeners were determined from the same marine species by Wright and co-workers in 2004.[7] A comparison of the antitumor activities of these marine polyketides revealed the crucial role of the ester chain. The potent biological activity[8] and intriguing chemical structure attracted many synthetic groups to embark on the total synthesis of lasonolides.[9,10] It is particularly noteworthy that the Trost group described a 16-LLS (longest linear sequence, LLS) synthesis of lasonolide A featuring a remarkable Ru-catalyzed alkene-alkyne coupling in a highly convergent manner.[9f,g] However, an unsatisfactory regioselectivity in the enyne coupling (the ratio of linear/branch: 2/1—3/1) might be a challenge for scaling up the synthesis. With these observations in mind, we commenced to devise a novel strategy to produce lasonolide A that is amenable for derivatization for future medicinal studies.

Our retrosynthetic analysis is depicted in Scheme 2. The disconnection of the olefin at different sites leads to five readily accessible building blocks **6**—**10**, which could be generally prepared within 4—8 steps from commercially available materials. The unprecedented bissulfone **8** could be derived from subsequent Mitsunobu reactions of propan-1,3-diol with PTSH (5-mercapto-1-phenyl-1*H*-tetrazole) and BTSH (2-mercaptobenzothiazole). All of the olefination processes would produce the requisite alkenes in a stereodefined manner. The key element to this synthetic design is the construction of two tetrahydropyrans bearing four stereogenic centers. The method we relied on is an oxa-Michael cyclization under thermodynamic control to secure the *cis*-configuration of C7/C11 and C19/C23. Herein, we describe the full details of the synthesis of macrolide lasonolide A.[11]

Scheme 2 Retrosynthetic analysis of (–)-lasonolide A (**5**)



Results and Discussion

Syntheses of the two tetrahydropyran rings

Enantio-integration. The *hidden* symmetry within the two tetrahydropyrans A and B (**7** and **10**) permits us to devise a novel strategy to access such multifunctional subunits (Figure 2). We envisioned that both enantiomers of racemic allenic alcohol can be evolved to form the individual fragments (Scheme 3). Following our previous exploration on the construction of tetrahydropyrans,[3c] the quaternary carbon center in the A-ring could be derived from the hydroboration-allylation of *R*-**11**, while the acetate form of *S*-antipode **12** is engaged into the preparation of the B-ring (*vide infra*). However, the latter clearly requires an unprecedented sequential oxidation to install two hydroxy groups in a stereoselective manner. According to conventional enantioconvergent synthesis,[12] the *S*-antipode is likely to be converted into the same key intermediate derived from the *R*-enantiomer. Thus, in our case, the novel enantio-integrative approach can be considered to be a chirality-economic process.[13]

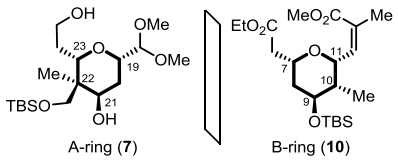
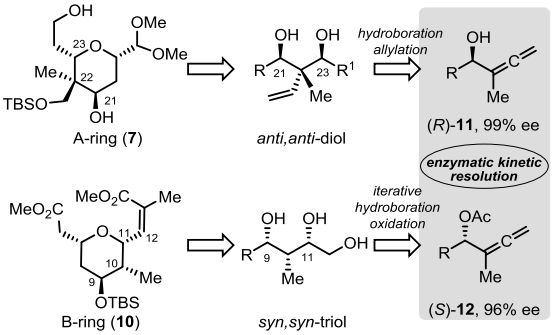


Figure 2 The *hidden* symmetry between **7** and **10**.

Scheme 3 The chirality-economic enantio-integrative approach



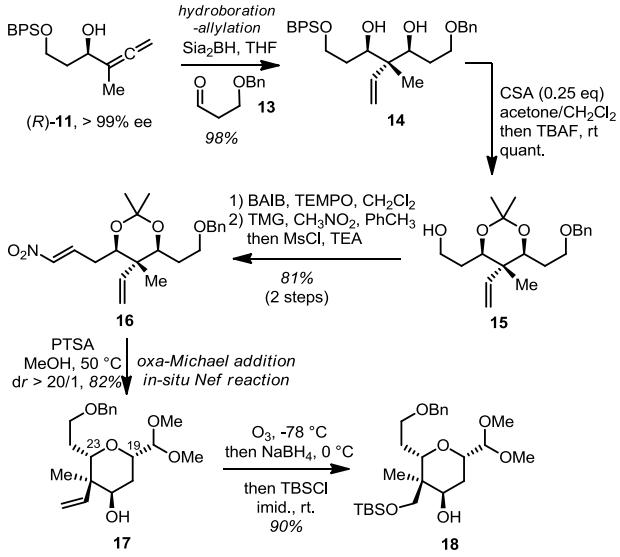
A-ring synthesis. Based on our previous work,[3c] a similar route toward the A-ring could be devised with oxa-Michael cyclization as a key transformation; however, the additional carbon chain derived from the Michael addition of the α,β-unsaturated ester (as **3** shown in Scheme 1) has to be adjusted to furnish the double bond between C17—C18 (Scheme 2). Therefore, a nitro-olefin was proposed as an alternative Michael acceptor, because an aldehyde can be exposed from the nitro group via the Nef reaction. A challenge to implement this functional group transformation is that, under basic or acidic conditions, the Nef reaction may induce a severe epimerization of the chirality at the α-position of the aldehyde. Little was known about the feasibility of using such a strategy in the THP-ring formation with high diastereoselectivity.[14] The only tractable case was reported by the Petruš group, in which a pair of 1/1 diastereoisomers was formed after a cascade process that included the oxa-Michael addition of the nitro-olefin and Nef reaction to form the acetal group.[15]

The synthesis commenced from allenic alcohol **11**, a highly optically pure (>99% ee) building block derived from an enzymatic kinetic resolution by lipase AK.[16] The recovered *R*-enantiomer of allenic alcohol was first subjected to hydroboration with disiamylborane (Sia_2BH). The resulting *Z*-allylborane further reacted with aldehyde **13** via a kinetic control to give *anti*,*anti*-1,3-diol **14** as a single stereoisomer (Scheme 4). The configuration of the 1,3-diol was established by Rychnovsky's method[17] and was consistent with the requisite stereochemistry in the A-ring. Consequently, masking the 1,3-diol in the presence of camphorsulfonic acid (CSA) and subsequently removing the *tert*-butyldiphenylsilyl (BPS) group with TBAF produced a quantitative yield of **15** bearing a primary alcohol. After treatment with 2,2,6,6-tetramethylpiperidinyloxy (TEMPO) and bis(acetoxy)iodobenzene (BAIB),[18] the resultant aldehyde was immediately subjected to the Henry reaction to react with nitromethane in the presence of tetramethylguanidine (TMG). The subsequent triethylamine-promoted elimination of the corresponding mesylate[19] afforded the requisite *E*-nitro olefin **16** in a combined 81% yield. This one-pot protocol was operationally simple and a high yield was achieved on

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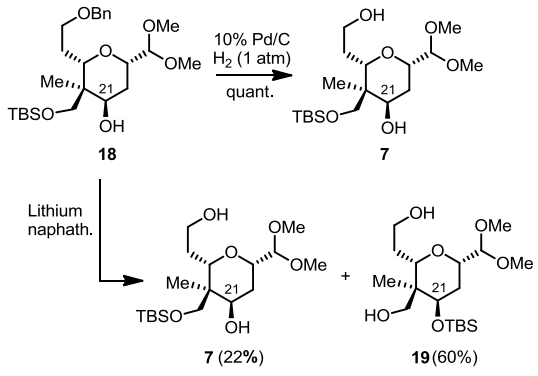
gram-scale. The standard condition using *para*-methylphenylsulfonic acid (PTSA) to promote the cyclization was remarkable. We found a cascade process, including deketalization, oxa-Michael cyclization, the Nef reaction, and acetalation of aldehyde in a one-pot protocol to result in tetrahydropyran **17** with excellent diastereoselectivity (*ds* >20/1) and isolated yield (82%). The mild conditions to realize the Nef reaction was attributed to a thermodynamically favorable formation of the acetal product **17**. The extensive NOE studies confirmed the *cis*-configuration of the stereochemistry between C19 and C23. At this stage, the vinyl group was readily converted to the hydroxymethyl group through ozonolysis and immediate reduction with $NaBH_4$. Without further purification, the crude mixture was suitable for the subsequent selective silylation to form **18** in 90% yield.

Scheme 4 Preparation of the A-ring acetal **18**



The following installation of the ester chain would be convenient if the benzyl group is removed. Indeed, debenzylation was readily performed under heterogeneous hydrogenation to give the primary alcohol **7** with excellent yield (Scheme 5). We challenged ourselves to realize such a transformation without transition metal catalysis. A transition-metal free synthesis would be particularly intriguing in terms of eliminating a tedious purification process in the complex molecule synthesis, especially for those intermediates or the final product that are in an oil form and not purifiable via recrystallization.[20] A lithium-naphthalene system[21] was adopted for debenzylation in presence of the silyl group. To our surprise, a silyl-migrated product **19** (60% yield)

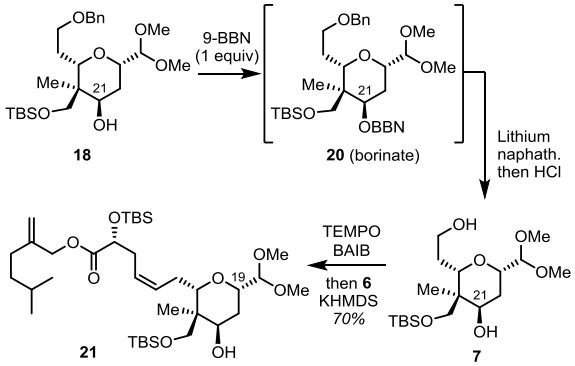
Scheme 5 Debenzylation and silyl-migration product **19**



caused contamination during the process. A single-crystal analysis revealed the structure of **19**, in which the silyl group was relocated to the secondary alcohol at C21.

A common solution to suppress the interference of the proximal C21-OH is the introduction of another protecting group. However, the additional protection-deprotection steps will dramatically decrease the synthetic efficacy.[22] The stability of borinate (the boron-ate complex) generally encountered in the Brown hydroboration-oxidation[23] inspired us to propose borane as a traceless protecting group. With the consideration of steric hindrance and stability, 9-BBN was defined as the optimal reagent of choice. Compound **18** was first treated with 9-BBN and then subjected to debenzylation by lithium naphthalene (Scheme 6). To our delight, a clean transformation occurred and the desired product **7** was isolated with 91% yield. Consequently, the primary alcohol **7** was subjected to a selective oxidative and immediate Wittig reaction with phosphonium salt **6**, prepared at the gram-scale by Trost's protocol[9f] to furnish the top fragment **21** at 70% isolated yield. The tolerance of the borinate intermediate to lithium-naphthalene would be valuable in other chemical transformations. A recent interesting case reported by the Hoveyda group also revealed this valuable approach (protection in a borinate form) in Mo- or Ru-based ring-closing metathesis (RCM).[24]

Scheme 6 Preparation of the top fragment **21**



B-ring synthesis. Although there is no quaternary carbon center in the B-ring, the installation of two hydroxy groups is still a difficult task, especially to achieve a good facial selectivity of the hydroxylation of the trisubstituted acyclic alkene (Figure 3). Brown and co-workers reported an excellent stereoselectivity of the hydroboration-oxidation of the cyclic trisubstituted alkene.[25] For the acyclic variants, hydroboration is generally slow and often suffers from competing alkene isomerization. Moreover, it is also difficult to achieve excellent stereoselectivity due to an $A^{1,3}$ strain.[26] Based on the observations in the various precedents, the hydroboration of the *E*-trisubstituted alkene mainly resulted in *anti*,*anti* or *syn*,*anti*-diastereomers of 1,3-diols.[27] In a recent solution reported by the Evans group, the hydroboration of *Z*-alkene with $ThexBH_2$ via a bis-alkoxysilane medium 7-membered ring resulted in *syn*,*syn*-1,3-triols with excellent diastereoselectivity (*ds* ≥19 : 1).[28] We thus envisioned that a possible boron-ate complex to chelate 1,4-diol may also promote the hydroboration of *Z*-alkene to favor a *syn*,*syn*-isomer through

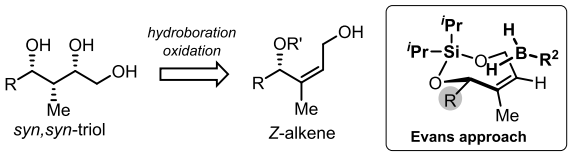


Figure 3 Synthesis of *syn*,*syn*-triol and Evans approach.

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mimicking the conformation of the silyl-templated medium ring (the chelation model in Figure 3).[29]

At the outset, a common challenge for the hydroboration of trisubstituted alkene was a severe double bond isomerization and migration that would scramble the hydroboration/oxidation to give a mixture of alcohols.[29] $ThexBH_2$ was selected as the reagent due to its superior reactivity for hydroboration of sterically bulky alkenes.[30] Formation of borinate complex and hydroboration with $ThexBH_2$ resulted in the desired triol **23a** albeit in an unsatisfied diasetereoselectivity (*ds* 1.4/1) along with the formation of *syn*-diol **23c**[31] (entry 1, Table 1). To improve the selectivity in hydroboration, a more sterically demanding Sia_2BH may provide a confining ate complex, which therefore facilitates the facial selectivity of hydroboration with $ThexBH_2$. However, this combination remained to show a severe isomerization of alkene (entries 2 and 3, Table 1). Application of a less reactive, and more selective 9-BBN allowed the subsequent hydroboration occur to deliver the triol favoring *syn*,*syn*-isomer (*ds* 2.5/1) (entry 4). Interestingly, decreasing the amount of 9-BBN further improved the diastereoselectivity (*ds* 4/1) to give *syn,syn*-triol **23a** in 61% isolated yield (entry 5). The inherent chirality in the borane reagent would affect the stereoselectivity as well as the conversion (entries 6—8) and identified the optimal conditions of using (+)-$IpcBH_2$ for the hydroboration with a synthetically useful yield (76%) (entry 7). Although the antipode (–)-$IpcBH_2$ resulted in a similar diastereoselectivity (*ds* 6.2/1) albeit the slower reaction rate (entry 8), the single borane reagent did not offer a better selectivity (entry 9).

In our previous work, an acyclic model (Felkin-Ahn-like model) was preferentially suggested to decipher the stereochemical outcome.[11] However, such model remains incomplete to understand the deterioration in diastereoselectivity when different sterically demanding boranes were applied for the hydroboration. Here, we would re-evaluate a chelation model (Figure 4). Interestingly, the conventionally adapted Houk model[30e,f] preferred a σ-donor (*i.e.*, the R group instead of OH or OR) to be antiperiplanar to the approaching borane and thus the hydroboration and subsequent oxidation will form another diastereoisomer, *anti*,*syn*-triol (Figure 4A). This interpretation is clearly distinct to the current result. We noticed that a smooth reaction of borane with alcohol occurred when 9-BBN[32] and $IpcBH_2$ were added subsequently. It was thus concluded that the borinate complex should be involved in the subsequent hydroboration. This notion was actually pointed out in the original Still's work, in which the formation of borinate (alkoxylborane) was found to be more rapidly than the hydroboration of alkene.[30c] However, a simple modification of the Houk model bearing the two borinate motifs would again deliver the *anti*,*syn*-isomer (Figure 4B). With the above information in mind, a chelation model, derived from the dehydrocoupling of 9-BBN with the primary allylic alcohol and the additional interaction with the sterically demanding secondary allylic alcohol, is thus assumed to act (Figure 4C). In this conformation of the borinate complex, the sterically bulky R group is pseudo-equatorial, facilitating the addition of $IpcBH_2$ to the least hindered and convex face of the trisubstituted alkene and thus to afford *syn*,*syn*-triol **23a** as the major product. Moreover, an increase of 9-BBN (>1.5

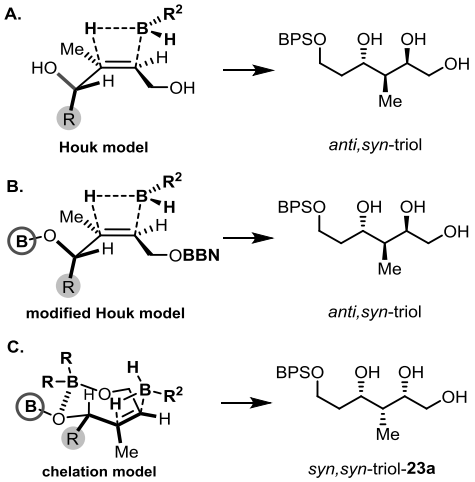
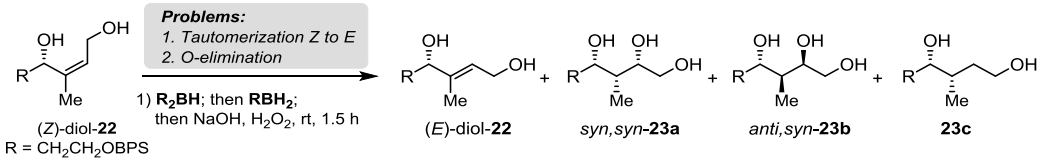


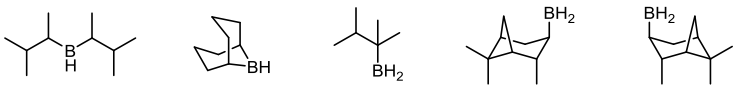
Figure 4 Comparison of the Houk model (A and B) with the current chelation model (C) for the formation of *anti*,*syn*-triol and *syn*,*syn*-triol **23a**.

Table 1 Optimization for the preparation of *syn*,*syn*-triol **12**[a]



entry[b]	Hydroxy protection		Hydroboration		conv.[c]/%	ratio *Z*/*E*	*ds*[d]	yield (**23a**)[e]/%
	R_2BH	*T*/°C, *t*/h	**RBH_2**	*T*/°C, *t*/h				
1	$ThexBH_2$ (1.1)	–78 to 0, 1 h	$ThexBH_2$ (3.5)	0, 2 h	>95	1/6	1.4/1	—[f]
2	Sia_2BH (2.0)	–78 to –30, 1 h	$ThexBH_2$ (3.5)	–30 to 0, 4 h	—	1/2	—	—
3	Sia_2BH (1.0)	–78 to –30, 1 h	$ThexBH_2$ (3.5)	–30 to 0, 4 h	—	1/7	—	—
4	9-BBN (2.1)	–30 to –10, 1 h	$ThexBH_2$ (3.5)	–10 to 0, 4 h	>95	—	2.5/1	—
5	9-BBN (1.1)	–30 to –10, 1 h	$ThexBH_2$ (3.5)	–10 to 0, 4 h	>95	—	4.0/1	63
6	9-BBN (1.1)	–30 to 0, 1 h	(+)-$IpcBH_2$ (3.5)	–10, 2 h	>95	—	5.0/1	70
7	9-BBN (1.1)	–30 to –10, 1 h	(+)-$IpcBH_2$ (3.5)	–20, 2 h	>95	—	6.4/1	76
8	9-BBN (1.1)	–30 to –10, 1 h	(–)-$IpcBH_2$ (3.5)	0, 5 h	76	—	6.2/1	—
9[g]	(+)-$IpcBH_2$ (4.4 equiv), –78 to –30, 3h				>95	—	4.1/1	—

structures of boranes used in this work



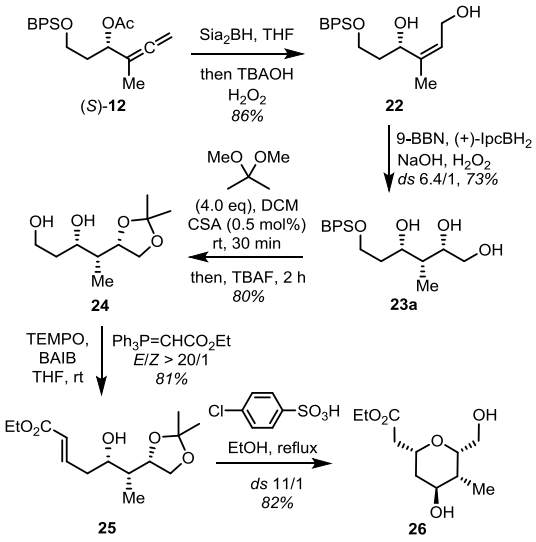
[a] Reaction conditions: (*Z*)-diol-**22** (150—200 mg), THF (1.0 mL); [b] The substrate was *rac*-(*Z*)-diol-**22** in entries 1—5; the substrate was (*S*)-(*Z*)-diol-**22** (>99% ee) in entries 6—9; [c] The conversion was determined by ^1H NMR (400 MHz, $CDCl_3$); [d] Diastereoselectivity (*ds*) = *syn*,*syn*-**23a**/*anti*,*syn*-**23b**; [e] Isolated yield; *syn*-**23c** < 5% in all entries (determined by ^1H NMR); [f] The ratio of Σ(**23a**+**23b**)/**23c** was 2.4/1 in entry 1 while the minor diol **23c** was detected with a trace amount in other entries; [g] The reaction was performed at –30 °C for 3 h.

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equiv) resulting in a lower diastereoselectivity (*ds* 2.7/1) indicated that 9-BBN was inferior in terms of the reactivity and diastereoselectivity with comparison to $ThexBH_2$ and $IpcBH_2$ and that was known in Still's work.[30c] It is intriguing that the diastereoselectivity reversed by introducing a secondary hydroxy group clearly requires more mechanistic investigation as well as its potential application in the stereoselective synthesis of polyols.

Following the optimal condition, 2-*Z*-Ene-1,4-diol **22** first reacted with 9-BBN (1 equiv) and the proposed boron-ate complex was then subjected to another borane reagent for the hydroboration ($IpcBH_2$, 3.3 equiv in THF). The corresponding triol **23a** can be obtained in 73% yield and good diastereoselectivity (*ds* 6.4/1) (Scheme 7).[33] The reaction was readily performed in gram-scale and a sufficient material was thus accumulated.

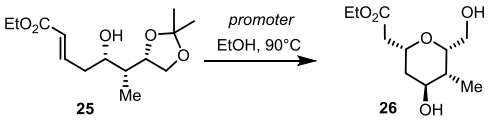
Scheme 7 Preparation of the B-ring **26**



It is noteworthy that the optical purity of the corresponding 1,4-diol **22** was further enriched up to 99% ee from 95.5% ee after recrystallization. It is also important to improve the optical purity of the final polyketide target, because most of the intermediates are in oil form in total synthesis. The following acid-promoted protection of the vicinal diol and subsequent removal of the BPS group by TBAF yielded diol **24** in 82% yield. The oxidative Wittig reaction secured the Michael acceptor, which allowed the next oxa-1,4-addition to form the *cis*-tetrahydropyran B-ring **26** with high diastereoselectivity (*ds* 11/1) and 82% yield (Table 2). Due to the lack of the Thorpe-Ingold effect[34] as that in compound **16** (Schemes 4), a stronger acid as promoter was required to promote the requisite cyclization with a good yield and high stereoselectivity. PCSA (*para*-chlorobenzenesulfonic acid) was effective to promote the cyclization with a moderate yield (entry 3). Further increasing the acidity did not offer a better outcome (entries 4 and 5). The optimal condition to deliver THP **26** was using 2 equivalents of PCSA at reflux in EtOH (entry 6).

Although the standard oxidative Wittig reaction could convert diol **26** to a conjugated ester, the severe epimerization at C11 implicated that the delicate control of the reaction conditions would be necessary (entry 1 in Table 3). Therefore, a series of TEMPO derivatives, oxidants, and other reaction parameters were screened (Table 2). A direct oxidation with DMP did not afford any desired product (entry 2). A mild oxidant, $TEMPOBF_4$, was effective to suppress the epimerization (entry 3). By using strong oxidant, TCCA,[35] a faster reaction occurred and the ratio was improved to 4/1 albeit with a low isolated yield (entry 4). We then focused on TEMPO-derivatives. In THF, substitution at C4 (**27a** and **27b**) delivered a higher ratio albeit a sluggish reaction (entries 5 and 6). The less polar solvent (*i.e.*, CH_2Cl_2) dramatically accelerated the reaction (entries 8 and 9). 4-Benzyloxy-TEMPO (**27c**)[36,37] was found to be ideal to promote the formation of aldehyde with a good selectivity at C11 (*S*/*R*: 10/1) (entry 10) before the subsequent HWE reaction.

Table 2 Optimization of the cyclization of **25**



entry	promoter (equiv)	*T*, *t*	yield (**26**)[a]/%	*ds*[b]
1	PTSA (1.2)	50 °C, 12 h	N.R.[c]	—
2	PCSA (1.2)	50 °C, 10 h	<10	—
3	PCSA (2.0)	80 °C, 60 h	69	8/1
4	PNSA (2.0)	80 °C, 85 h	67	7/1
5	TCSA (2.0)	80 °C, 85 h	59	8/1
6	PCSA (2.0)	reflux, 70 h	82	11/1
7	PCSA (4.0)	reflux, 36 h	59	11/1

[a] The isolated yield of **26**; [b] Diastereoselectivity (*ds*); [c] No reaction.

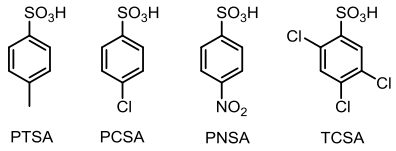
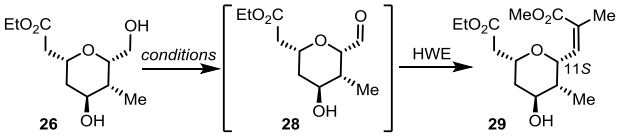
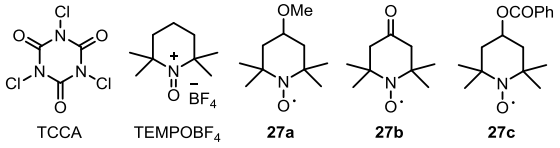


Table 3 Optimization of the oxidation of **26**



entry	conditions (equiv)	time	conv./%	**29** (11*S*/11*R*)
1	TEMPO (0.1), BAIB (1.2)	4 h	>95	1.7/1
2	DMP (1.2)	5 h	>95	complex
3	$TEMPOBF_4$ (2.6)	29 h	>95	3/1
4	TEMPO (0.1), TCCA (0.75)	39 min	>95	4.4/1[a]
5[b]	**27a** (0.75), BAIB (1.3)	26 h	<7	8/1
6[b]	**27b** (0.75), BAIB (1.3)	17 h	15	>10/1
7[b]	**27c** (0.75), BAIB (1.3)	26 h	N.R.[d]	—
8[c]	**27a** (0.25), BAIB (1.3)	8 h	80	5/1
9[c]	**27b** (0.25), BAIB (1.3)	8 h	42	10/1
10[c]	**27c** (0.25), BAIB (1.3)	8 h	>90	10/1

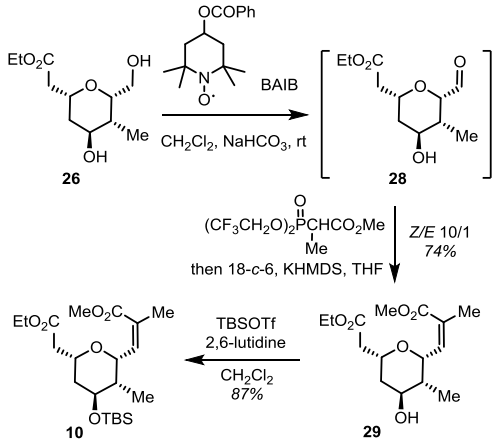
[a] The combined yield of **29** was 35%; [b] THF as solvent at room temperature; [c] CH_2Cl_2 as solvent at room temperature; [d] No reaction.



The optimized conditions were successfully applied in a gram-scale synthesis of *Z*-olefin **29** in 74% isolated yield (Scheme 8). To avoid the possible interference of the free hydroxy group at C9 in the following transformations, compound **29** was then subjected to protection with a TBS group to furnish diester **10**.

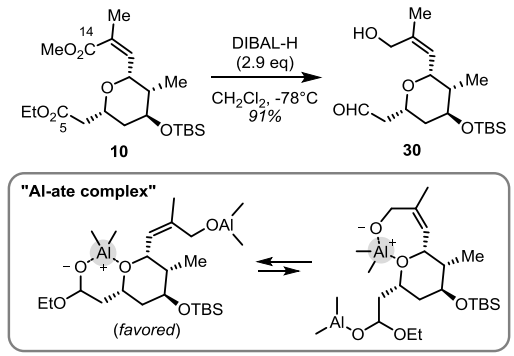
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Scheme 8 Preparation of the modular **10**



The reactivity difference between the conjugated ester at C14 and the isolated ester at C5 may provide enough space for a chemoselective reduction. Indeed, when 2.9 equivalents of DIBAL-H were introduced, the conjugated methyl ester was preferentially reduced to allylic alcohol, while the isolated ethyl ester remained at the aldehyde stage. The resulting **30** was isolated with a remarkable 91% yield (Scheme 9). The excellent regioselectivity may be attributed to a favorable six-membered chelation, which forms a stable Al-ate complex and thus suppresses the further reduction of hemiacetal to alcohol, while the conjugated ester is completely reduced to the allylic alcohol.

Scheme 9 Selective reduction of diester **10**

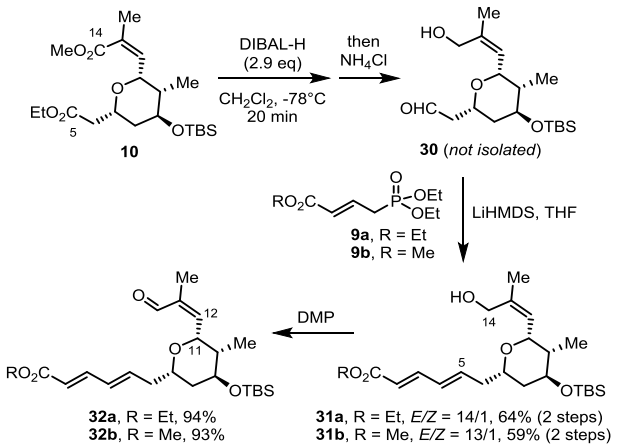


With a sufficient amount of **30** in hand, we then decided to introduce the C2-C5 dienone motif via an HWE elongation. It is also interesting to find that the crude product **30** from the reduction was suitable for the subsequent HWE reaction with phosphonate **9a** and **9b**[38] to form diene-ethyl ester **31a** and diene-methyl ester **31b** with good yield and excellent *E*,*E*-stereoselectivity (≥13/1) (Scheme 10). The different esters **9a**/**9b** were designed for the future hydrolysis (*vida infra*, Scheme 14). The oxidation of allylic alcohol by Dess–Martin periodinane (DMP)[39] proceeded well to deliver enal **32** at a high yield without the deterioration of the *Z*-C12-alkene configuration and the stereochemistry at C11.

Fragment coupling

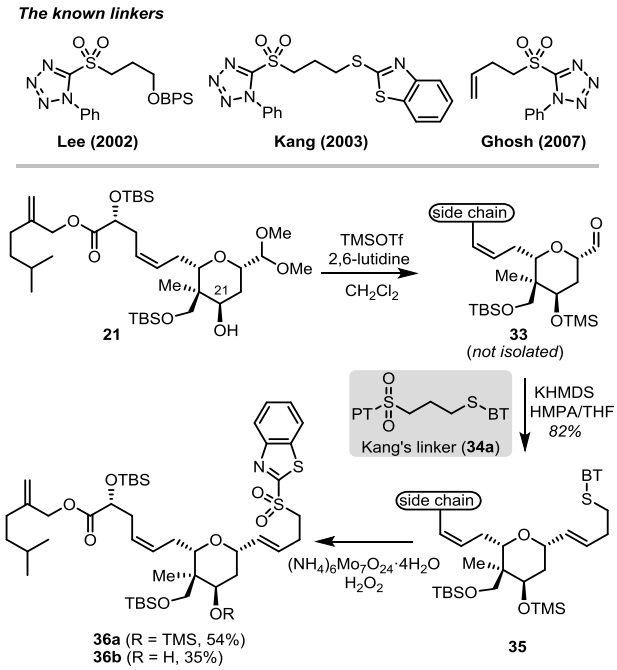
With the successful construction of the two complex fragments **21** and **32**, the final installation of the skipped diene linker was executed to complete the construction of the whole carbon framework. The Lee group first introduced an iterative Julia-Kocienski olefination process (repeating a three-step protocol: thiolation-oxidation-Julia reaction), which became a standard strategy in the following synthetic efforts.[6a] Kang and coworkers

Scheme 10 Preparation of the dienoic ester **32**



introduced a sulfone-thiol ether reagent, which could omit the Mitsunobu reaction to install the BT-thiol ether on a complex substance (Scheme 11).[9a] The Ghosh group synthesized a homoallylic BT-sulfone to install the same functional motif through the cross-metathesis method.[9d] The first Julia reaction was not applied; however, additional steps were required to install the terminal alkene (C18-C19) prior to the requisite cross metathesis (CM) reaction.

Scheme 11 Preparation of the BT-sulfones **36a** and **36b**

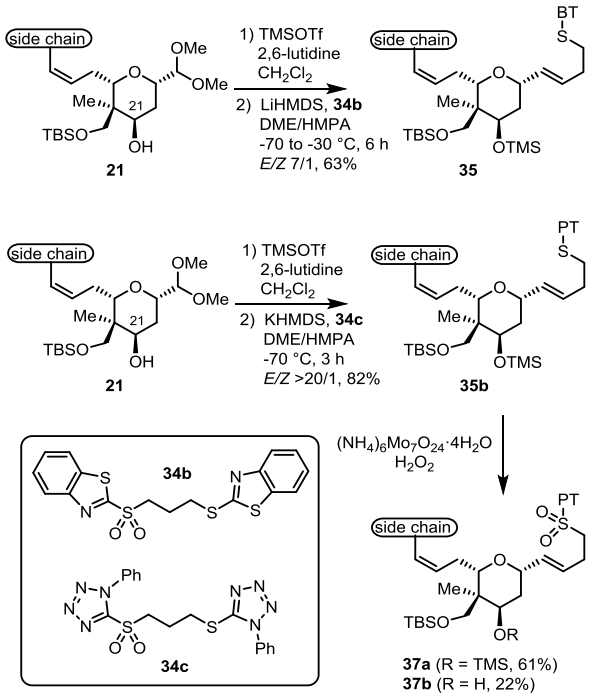


To install the skipped diene (C14-C15 and C17-C18) via the Julia reaction, it was found to be extremely difficult to convert the acetal **21** into aldehyde **33** due to sensitive functional groups and epimerization at C19 under acidic conditions (Scheme 11). To our delight, a modification of Kita's condition[40] successfully resulted in the requisite aldehyde **33** at C18 along with the protection of the secondary alcohol at C21 with a trimethylsilyl group. This compound can be used for the next Julia-Kocienski olefination with Kang's linker (**34a**) in the presence of KHMDS to deliver *E*-alkene **35** in 82% yield with a high stereoselectivity (*E*/*Z* >20/1). After oxidation, the BT-sulfone derivatives **36a** and **36b** were prepared in 54% and 35% yield, respectively. We also synthetized

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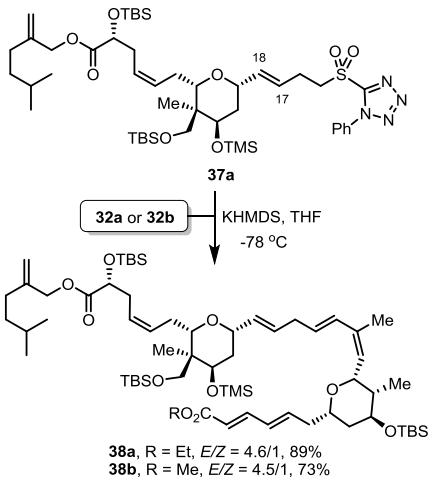
other linkers (**34b** and **34c**) for this chain elongation (Scheme 12). The PT-sulfone **34c** offered a good stereoselectivity (*E/Z* 11/1) for the first Julia olefination, while the BT-sulfone **34b** gave **35** in a lower *E/Z* ratio (11/1 for **34c** and 7/1 for **34b**). The following oxidation of thioether with ammonium heptamolybdate to sulfone[41] proceeded smoothly to yield PT-sulfone **37a** if the work-up procedure was neutral. However, the TMS group was fragile and easily removed under oxidative conditions. The corresponding **37a** (61% yield) or **37b** (22% yield) can be isolated under optimized conditions.

Scheme 12 Sulfones **34b** and **34c** in Julia/Julia-Kocienski olefination



Before we systematically evaluated the two sulfones **36** and **37** for the second Julia reaction, we first chose **37a** to react with KHMDS in THF, which then immediately reacted with aldehyde **32** to deliver the polyene structure **38a** with 89% yield and a moderate ratio of *E/Z* (4.6/1 for C14/C15) (Scheme 13). The methyl ester **38b** was also obtained in a similar yield (73%) and stereoselectivity (*E/Z* 4.5/1). It would be optimistic to further process the final target after hydrolysis and macrolactonization as the downstream steps.

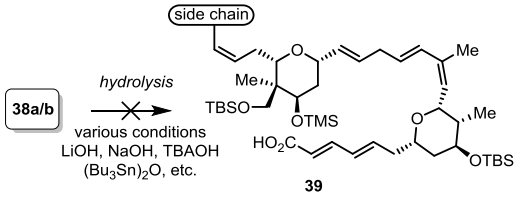
Scheme 13 Preparation of **38a** and **38b** by the second Julia-Kocienski olefination



Dead-End and Detour: 9-BBN as a traceless protecting group for Julia olefination

The hydrolysis problem. To our surprise, the hydrolysis of the ethyl or methyl esters to a seco-acid **39** turned out to be problematic (Scheme 14). Instead of the dienoic methyl ester, the ester chain was hydrolyzed under various hydrolysis conditions we examined. It is clear that the conjugation (C2-C5) attenuates the reactivity of the methyl ester and thus renders **38** resistance to hydrolysis. An alternative deconjugation of **38a/b** through a thio-Michael addition may promote hydrolysis at the requisite site, but the complex stereochemistry and regioselectivity and the additional step of oxidative elimination render such an approach impractical.

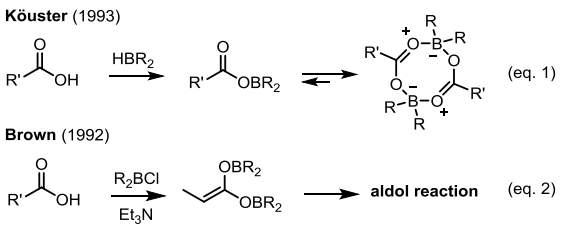
Scheme 14 The hydrolysis challenge



The hydrolysis challenge definitely requires a detour to liberate the free acid for macrolactonization. We anticipated the use of other facile esters for the late-stage hydrolysis would be practical. It requires the HWE reagent (*i.e.*, **9**) to be redesigned for olefination. Moreover, this functional group must be tolerable to the previous HWE and Julia olefination conditions. However, the sensitivity of various functional groups in **36** or **37** may make this approach unfeasible. A second approach is carrying out the Julia olefination with a free seco-acid. The latter was even more demanding. A strong base used for the Julia reaction would be quenched by a free acid (*vide infra*).

During the synthesis of **7** (Scheme 6), 9-BBN was applied as a traceless protecting group for the secondary alcohol to tolerate the basic conditions. It is well-known that a free acid could readily react with borane to form various boron-ate complexes.[42] The direct boronation of carboxylic acid was reported in Köuster's work (eq. 1) (Scheme 15). Extensive studies with the NMR, FT-IR, and X-ray analyses verified the existence of the borinate complex as a mixture of monomer and dimer forms. Moreover, Brown and coworkers reported an indirect approach to synthesize borinate from carboxylic acid with dicyclohexylchloroborane (Chx_2BCl) in the presence of triethylamine (eq. 2),[43] in which vinyloxyborane was used for aldol reaction. However, this information has not been spotlighted and the chemical reactivity of carboxyl borinate remains to be demonstrated.

Scheme 15 Borane-acid ate complex (R = alkyl)

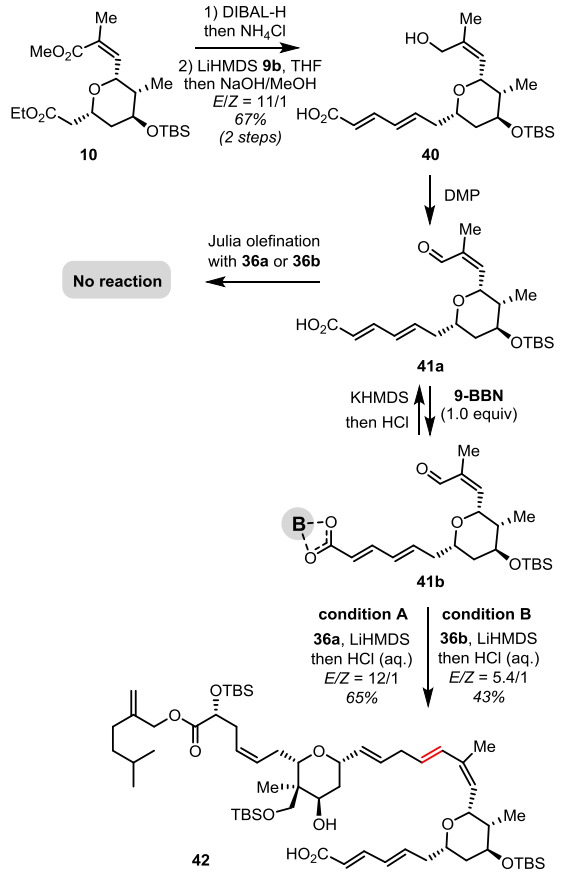


We envisioned that such a borinate complex would be stable enough to survive under basic conditions if no additional strong coordination ligands to boron (such as the alkoxide or hydroxy

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anion) are introduced.[44] To evaluate this assumption, aldehyde **30** was subjected to chain elongation with phosphonate **9b** and immediate hydrolysis to deliver dienoic acid **40** in 83% yield (Scheme 16). After oxidation, the seco-acid **41** was obtained with good yield. It was not surprising that the direct Julia reaction of **41** with BT-sulfone **36a**/**b** did not produce any of the desired coupling products. To mimic the basic conditions of the Julia reaction, acid **41a** was pretreated with 1 equivalent of 9-BBN, and the proposed borinate intermediate **41b** was mixed with 4—6 equivalents of KHMDS at –45 °C and then warmed up to room temperature for 30 min. After being quenched with aqueous HCl, the free acid **41a** can be fully recovered (>90% yield) without appreciable epimerization. Moreover, a single-crystal analysis of acid **40** unambiguously established the stereochemistry of the stereogenic center on the tetrahydropyran ring and three alkene geometries. Accordingly, the Julia reaction was resumed. The BT-sulfone **36a** was deprotonated with LiHMDS and then reacted with the in situ generated borinate complex **41b**. After the acidic work-up, the coupling product seco-acid **42** was obtained in 65% isolated yield with an excellent *E*/*Z* stereoselectivity (12/1 for C14-C15) (**condition A**), whereas using **36b** bearing a free C21-OH group resulted in a lower *E*/*Z* ratio (5.4/1) (**condition B**).

Scheme 16 The boron-ate complex as a traceless protecting group

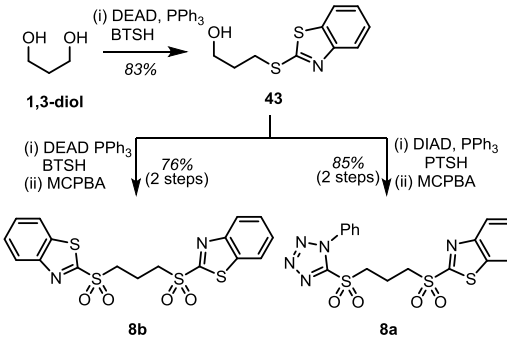


The complete synthesis

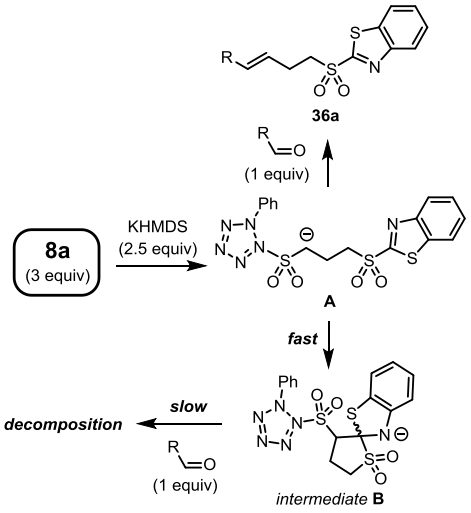
Double-headed sulfone. The successful Julia reaction with the tentatively protected acid **41b** encouraged us to further improve the overall efficiency of the total synthesis. We noticed that the sulfone was actually very sensitive to different bases. For instance, PT-sulfone is sensitive to KHMDS while the deprotonation of the BT-sulfone is generally realized by LiHMDS. We thus envisioned that a bissulfone reagent may be useful for the sequential Julia olefination of different aldehydes without the additional oxidation steps as shown in Scheme 11.

As a proof-of-concept, double-headed (DH) sulfones **8a** and **8b** bearing the BT and PT sulfones were prepared in three steps from propan-1,3-diol (Scheme 17).[45] The DH-sulfones were crystalline white solids and readily purified after recrystallization (>99% purity). Following the conventional procedure, after the deprotonation of bissulfone **8a** with KHMDS in DME/HMPA, the addition of aldehyde **33** did not result in any detectable amount of the desired product **36a** (entry 1, Table 4). Interestingly, when the order of addition was reversed and KHMDS was added after the pre-mixed **8a** and **33**, the first Julia reaction occurred to deliver **36a** with 73% isolated yield and a good *E*/*Z* selectivity (8.7/1) (entry 2). This protocol indicated the sensitivity of bissulfone to the strong base and the subsequent side reactions that would consume the reagent (*e.g.*, forming a spiro intermediate **B** which may be inert to the Julia reaction) (Scheme 18).

Scheme 17 Preparation of DH-sulfones **8a** and **8b**



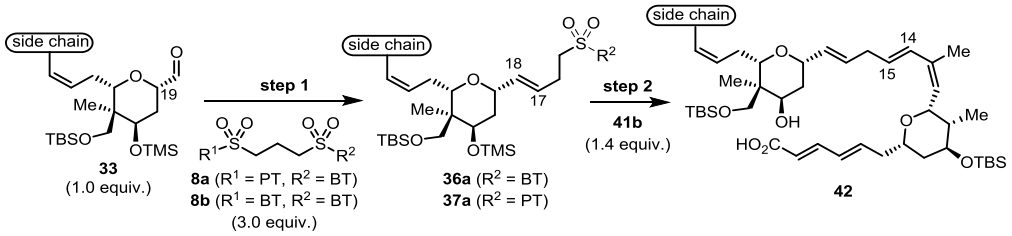
Scheme 18 A proposed degradation pathway of bissulfone **8a**



To further improve the stereoselectivity, the reaction parameters, such as the strong bases (KHMDS and LiHMDS), solvents (THF, DME, and DMF) and additives (HMPA and 18-*c*-6), were subsequently screened. Without the presence of HMPA, a lower stereoselectivity was observed (entry 3). Further addition of 18-*c*-6 and the replacement of DME with THF did not give better results (entries 4—6). A combination of DMF and HMPA resulted in an excellent selectivity favoring the *E*-alkene at C17-C18 (>20/1) in 83% isolated yield (entry 7). For bissulfone **8b** (bis-BT variant), KHMDS only resulted in a low conversion (44%) and stereoselectivity (*E*/*Z* 2.1/1) (entry 8). By switching to LiHMDS as the base, a slightly better *E*/*Z* ratio was achieved (3.9/1), but the conversion remained unsatisfactory (entry 9). The screening of

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Table 4 Optimization of the sequential Julia olefination (step 1)[a]



entry	base (equiv.)	sulfone	solvent	*T*/°C	conv./%	*E*/*Z*[b]	yield (**36a**)[c]
1[d]	KHMDS (2.5)	**8a**	DME/HMPA	–70	0	—	—
2	KHMDS (2.5)	**8a**	DME/HMPA	–70	>90	8.7/1	73%
3	KHMDS (2.5)	**8a**	DME	–70	>90	4.9/1	—[e]
4	KHMDS (2.5)	**8a**	THF/HMPA	–78	>90	5.0/1	—
5	KHMDS (2.5)	**8a**	DME/18-*c*-6	–70	>90	5.4/1	—
6	KHMDS (2.5)	**8a**	DME/HMPA/18-*c*-6	–70	>90	7.3/1	—
7	KHMDS (2.5)	**8a**	DMF/HMPA	–60	>90	>20/1	83%
8	KHMDS (3.0)	**8b**	THF/HMPA	–78	44	2.1/1	—
9	LiHMDS (3.0)	**8b**	THF/HMPA	–78	39	3.9/1	30%
10	LiHMDS (6.0)	**8b**	THF/HMPA	–78	<10	—	—
11	LiHMDS (3.0)	**8b**	DME/HMPA	–70	28	—	—
12	LiHMDS (3.0)	**8b**	DMF/DMPU	–70	<5	—	—

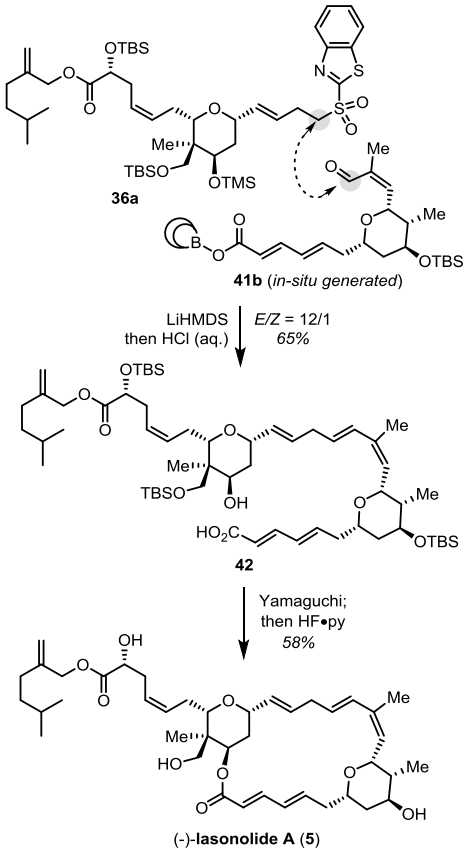
[a] The reaction was run on a 0.05 mmol-scale **33** (1 equiv.); the bissulfone and aldehyde were pre-mixed before the addition of the base except for the reaction in entry 1; [b] The *E*/*Z* ratio for the C17-C18 alkene and conversion (%) were determined by ^{1}H NMR (400 MHz, $CDCl_3$); [c] Isolated yield; [d] The aldehyde was added after the pre-mixed bissulfone and base; [e] Not performed.

the solvents did not result in improved yields and stereoselectivities (entries 10—12).

After optimizing the first Julia reaction of bissulfone **8a** and KHMDS as the base, we then turned to tackle the second transformation. LiHMDS was found to be effective for the deprotonation of BT-sulfone **36a** (derived from **8a**) and promoted the Julia olefination of borinate-aldehyde **41a** to deliver the requisite seco-acid **42** with a 63% isolated yield and excellent stereoselectivity (*E*/*Z* for C14-C15: 12/1) (entry 1, Table 5). The addition of HMPA dramatically decreased the stereoselectivity (6.7/1) (entry 2). PT-sulfone **37a** (derived from **34c**) was clearly not suitable because both the isolated yield and stereoselectivity were not satisfactory (entries 3—5). From these experimental executions, a bissulfone based on the two PT substituents was not further explored. Therefore, the optimal combination for the sequential Julia olefination would be a combination of the conditions in entry 7 of Table 4 and entry 1 of Table 5.

The protocol of the sequential Julia reaction was thus applied in the coupling of two complex fragments, namely, aldehydes **33** and **41b**, derived from acetals **21** and **41a**, respectively. After an acidic work-up, the desired acid **42** was isolated with 65% yield at a 200-mg scale (Scheme 19). With sufficient **42** in hand, Trost's protocol[9f] was adopted and the Yamaguchi esterification[46] with 2,4,6-trichlorobenzoyl chloride generated a macrolide that was immediately subjected to global desilylation to produce (–)-lasonolide A (**5**) in 58% isolated yield with spectroscopic data identical to those reported in the literature.[5a,9f] From the readily available racemic allenic alcohol, the total synthesis was completed in a 15-LLS sequence (12 telescoped steps) in 12% overall yield, which thus constitutes the shortest synthesis to date.

Scheme 19 The final steps for the total synthesis of (–)-lasonolide A (**5**)



Conclusions

We summarized here a modular synthesis of macrolide lasonolide A via a novel chirality-economic approach to integrate two tetrahydropyran rings in a stereoselective manner. The overall synthetic sequence is scalable and provides high diastereoselectivity for the most stereodefining steps. The access to chiral allenic alcohols based on a reliable enzymatic kinetic resolution is

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Table 5 Optimization of the sequential Julia olefination (step 2)[a]

entry	**36a**/**37a**	base (equiv.)	solvent	*T*/°C	*t*/h	conv.[d]/%	*E*/*Z*[b]	yield (**42**)[c]
1	**36a**	LiHMDS (2.0)	DME	–75	6.5	>95	12/1	63%
2	**36a**	LiHMDS (2.0)	DME/HMPA	–70	3	>95	6.7/1	—[e]
3	**37a**[d]	KHMDS (2.0)	THF	–78	2	>95	5.4/1	43%
4	**37a**[d]	KHMDS (2.0)	DME	–70	2	>95	3.7/1	—
5	**37a**[d]	LiHMDS (2.0)	DME	–70	3	>95	5.1/1	—

[a] The reaction was run on a 0.05 mmol-scale **36a**/**37a** (1 equiv.); [b] the *E*/*Z* ratio for the C14-C15 alkene and conversion (%) were determined by ^{1}H NMR (400 MHz, $CDCl_3$); the reaction was worked up by treatment of HCl (aq. 1 N) after the second Julia reaction; [c] Isolated yield; [d] The sulfone **37a** was prepared from a two-step sequence (Scheme 12); [e] Not performed.

noteworthy and products of high optical purity can be conveniently accumulated at a decagram scale. The unprecedented bis-sulfone (DH-sulfone) designed for the sequential Julia olefination dramatically improves the synthetic efficacy of the intergration of a skipped diene.[47] Moreover, borane reagents were chosen as traceless protecting groups for the alcohols and dienoic acid to reduce the number of protecting-deprotecting steps. An additional feature of our synthesis is that no precious metal catalysts were used in the entire sequence, which will be advantageous for the development of future processes in which the omission of transition-metals removal shall be cost-effective. The modular synthetic route described here may support the material supply of lasonolide A due to the challenge of re-harvesting at this stage.

Experimental

The general procedure and all the characterization data of the synthetic intermediates as well as the procedures to prepare are listed in the Supporting Materials online. The known compounds were recorded in the previous report.[11a]

Supporting Information

The supporting information for this article is available on the WWW under https://doi.org/10.1002/cjoc.202000026.

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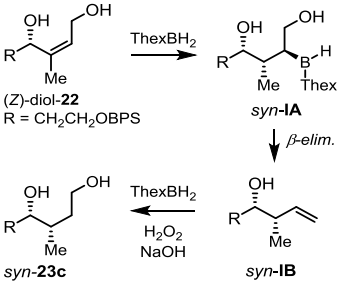
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[46] (a) Inanaga, J.; Hirata, K.; Saeki, H.; Katsuki, T.; Yamaguchi, M. A Rapid Esterification by Means of Mixed Anhydride and Its Application to Large-ring Lactonization. *Bull. Chem. Soc. Jpn.* **1979**, *52*, 1989–1993; For recent reviews, see: (b) Parenty, A.; Moreau, X.; Campagne, J.-M. Macrolactonizations in the Total Synthesis of Natural Products. *Chem. Rev.* **2006**, *37*, 911–939; (c) Parenty, A.; Moreau, X.; Niel, G.; Campagne, J.-M. Update 1 of: Macrolactonizations in the Total Synthesis of Natural Products. *Chem. Rev.* **2013**, *113*, PR1–PR40.

[47] For a very recent approach to access skipped dienes, see: Nagashima, Y.; Okada, Y.; Sato, T.; Chida, N. Enantioselective Stereodivergent Approach to α-Hydroxy Skipped Dienes: Synthesis of the Western Polyene Fragment of Corallopyronin A. *Chem. Lett.* **2019**, *48*, 1519–1521, and references cited therein.

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